COURSE TITLE : PAPER MAKING TECHNOLOGY
COURSE CODE : 5075
COURSE CATEGORY : E
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 52
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIOD
1	Understand wood Defibration, pulp storing , beating and refining	13
2	Understand Sizing, filling and loading of paper the function, types and parts of Head box and Wire part of paper machine	13
3	Understand Pressing, Drying, Calendering, Finishing, Coating of Paper.	13
4	Understand Chemical Recovery, Effluent Treatment, Air Pollution Control	13
TOTAL		**52**

COURSE OUTCOME :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	Understand wood Defibration, pulp storing , beating and refining
	2	Understand Sizing of paper Know the purpose of filling and loading Understand dyeing of paper Know the historical development of paper making
	3	Understand Pressing, Drying, Calendering, Finishing, Coating of Paper. Understand principles of drying Understand the coating operations on paper
2	4	Understand Chemical Recovery, Effluent Treatment, Air Pollution Control Understand caustisizing process Know effluent treatment

SPECIFIC OUTCOMES

MODULE I

1.1.0 Understand Defibration pulp storing beating and refining.

1.1.1 State defiberation stock preparation
1.1.2 Explain the equipment’s and operations of batch and continuous Hydrapulpers
1.1.3 Draw the figures of the equipment’s of Kneaders and deflakers and discuss their operations
1.1.4 State storage chest
1.1.5 Describe storage chest

1.1.6 Describe agitators with diagram
1.1.7 State the different types of stock and machine chests
1.1.8 Explain 2 types of storage chests with figures
1.1.9 State beating and refining process
1.1.10 State the theory of beating and refining process
1.1.11 Describe the factors influencing beating and refining
1.1.12 State Freeness and Fibrillation
1.1.13 Describe the effect of freeness and fibrillation on the properties of paper
1.1.14 List the different types of beaters & refuters
1.1.15 Explain different types of beaters & refiners such as – Hollander beater, conical refiners – narrow and wide angle, disc refiners – single and double disc
1.1.16 State consistency and stock proportioning
1.1.17 State the need of consistency controlling
1.1.18 List the different types of consistency controllers and stock proportions
1.1.19 State the working principles of consistency controllers and stock proportions
1.1.20 Classify properties of paper
1.1.21 Describe various properties according to the classification
1.1.22 Prepare sample surface properties
1.1.23 Explain methods of absorbency stock proportioning
1.1.24 List the types of absorbency stock proportioners
1.1.25 Describe various absorbency stock proportioners
1.1.26 State the cleaning and screening
1.1.27 Describe setting of centrifugal screens
1.1.28 Distinguish between the high density a low density cleaning
1.1.29 List the different types of screeners
1.1.30 Discuss pressure screens and centrifugal cleaners
1.1.31 Discuss sorters
1.1.32 Describe the theory of cleaning
1.1.33 Explain deculator with a neat diagram.
1.1.34 State the purpose of deaeration
1.1.35 Describe the working principles of deaerator with diagram
1.1.36 State the method of fiber recovery
1.1.37 List the different types of savealls
1.1.38 Compare the following types of saveall –settling and sedimentation. Conical saveall,
1.1.39 Explain flotation saveall and SP Krofta
1.1.40 State the need of filtration
1.1.41 Explain vacuum – disc filter

MODULE II

2.1.0 Understand Sizing, filling and loading of paper the function, types and parts of Head box and Wire part of paper machine

2.1.1 State sizing of paper
2.1.2 State the purpose of sizing
2.1.3 Differentiate between internal and external sizing
2.1.4 List the chemicals used for sizing such as – Rosin, Alum, Wax

2.1.5 Describe the preparation of above chemicals
2.1.6 State the theory of sizing
2.1.7 State and describe the parameters influencing sizing
2.1.8 Discuss different sizing mentors such as – Acidic, Alkaline and Neutral sizing
2.1.9 Discuss surface sizing methods and sizing chemicals
2.1.10 State the sizing agent – starch
2.1.11 Explain the figure – straight, Horizontal, Inclined and sped sizes (Presses)

2.2.0 State the purpose of filling and loading

2.2.1 List the types of fillers such as clay talc, titanium dioxide, calcium carbonate and magnesium carbonate.
2.2.2 Describe the above fillers and their chemical composition
2.2.3 Explain the chemical retention and effect of fillers on the properties of paper
2.2.4 Explain the theory of retention such as filtration co-floculation and mechanical attachment
2.2.5 List the properties of paper & Explain

2.3.0 State dyeing of paper

2.3.1 List the different types of dyes used in paper
2.3.2 Classify the various dyes and state their functions and uses
2.3.3 Describe the general properties of above groups of dyes
2.3.4 Describe fastness of dyes and different ways of adding the dye
2.3.5 State the penetration of dyes into the fiber
2.3.6 Explain the factors affecting the stock dyeing such as beating and refining sizing and filling pH order of addition and pulp characteristics
2.3.7 Describe wet strength additives
2.3.8 Explain the chemistry of additives
2.3.9 Discuss the influence of additives on cellulose
2.3.10 State the natural & synthetic types of additives
2.3.11 Describe the penetration and addition

2.4.0 Describe the historical development of paper making

2.4.1 State the function of head box
2.4.2 Describe the open and closed head box
2.4.3 Draw the head boxes and its parts such as distribution section, middle section, slice straight, converging head boxes
2.4.4 Define slices
2.4.5 Describe straight, convergent and combination slice with figures
2.4.6 Specify the relation between head of stock behind the slice and velocity
2.4.7 Calculate the velocity
2.4.8 State the role of distribution roll and high turbulence head box
2.4.9 Explain the different types of manifolds, head box over flow, head box recirculations, jet wire ratio and head calculations
2.4.10 Describe the general arrangement of paper machine systems
2.4.11 Describe the operation of paper machine systems and sheet formation

2.5.0 Explain the different types of paper machine wire part such as cylinder mould fourdrinier twin wire formers special formers and combination with diagrams

2.5.1 Explain the different components of wire part forming medium wire weave pattern and sheet formation.

MODULE III

3.1.0 Understand Pressing,Drying,Calendering,Finishing,Coating of Paper

3.1.1 State the features influencing drainage and sheet formation State the theory of pressing

3.1.2 State the different type of presses such as solid, suction, venta nip, reverse & trinip and give their uses

3.1.3 List the factors influencing dewatering at presses

3.1.4 Name the different types of felts such as woolen, synthetic, weaving and needling and state their need

3.2.0 State principles of drying

3.2.1 State different methods of drying such as conduction, convection and radiation

3.2.2 List the features affecting drying such as heat content temperature and moisture content and give the method of removing them

3.2.3 Draw the different type of dryers and its arrangements such as yankel cylinder (MG) multicylinder (MF) combined, infrared radiation randiant drying and state their functions

3.2.4 State the different types of steam and condensate the tial system such as cascade, pressure regulated, differential pressure regulated

3.2.5 Describe the different condensate removal methods such bucket scoop siphon rotating and stationary

3.2.6 List the different types dryer

3.2.7 State the principle of working of exhaust fans in the dryer

3.2.8 State the uses of permeability's cleaning & drying of dryer felts and screens

3.2.9 Describe the common features in drying and the method of removing them

3.2.10 Describe the operating problems related to the dryer section

3.2.11 Describe control of the dryer section

3.2.12 Specify the formula to calculate drying capacity

3.2.13 Specify the equation to calculate over all heat transfer coefficient and heat flux

3.2.14 Calculate the amount of heat transfer by using above equation (simple problems only)

3.3.0 Understand the coating operations on paper.

3.3.1 Describe coating operations

3.3.2 State advantages of coating

3.3.3 State coaters

3.3.4 List the coating chemicals and pigments

3.3.5 Describe coating chemicals and pigments

3.3.6 Describe the different types of coaters

3.3.7 State the theory of coating

3.3.8 Describe the different types of coaters, such as Brush coaters, cast coaters, blade coaters, knife coaters

3.4.0 State the theory of calendaring of calender rolls

3.4.1 State the function of super calenders and swimming rolls

3.4.2 Give the sketches of open and closed stock

3.4.3 State the importance of moisture on paper during calendaring

3.4.4 List the common calender troubles such as hard and soft spots sheet blackening cuts and scabs

3.4.5 Define the term reeling

3.4.6 State the need of different type of reels such as pope reels, core mind reels, friction or surface drive reels, four drum reels, two drum upright reels, three drum revolving reel

3.5.0 State finishing of paper

3.5.1 Explain the slitting & rewinding
3.5.2 Specify the winding troubles and their remedy
3.5.3 State the variables on calenders and winders
3.5.4 List the sheet cutting machines
3.5.5 Explain rotary (simplex and duplex) guillotine and their merits
3.5.6 State the need of sorting and counting of sheets
3.5.7 Describe the method of packing of reels and sheets
3.5.8 Describe roll wrapping machine with diagram
3.5.9 Describe the weighing & labeling
3.5.10 Define converting of paper
3.5.11 State corrugating of paper
3.5.12 Describe the uses of wet and dry waxing of paper
3.5.13 Describe the laminating of paper
3.5.14 List the properties sizes etc regarding paper bags and envelops
3.5.15 Explain the special characteristics of paper and board
3.5.16 State creeping of paper
3.5.17 State the news print
3.5.18 State the grease proof and glassine paper
3.5.19 Compare bond and currency papers
3.5.20 State the preparation method and uses of absorbent and water resistant paper
3.5.21 State Kraft paper and sack draft
3.5.22 State the method of making tissue paper for the toilet and cigarette

MODULE IV

4.1.0 Understand Chemical Recovery, Effluent Treatment, Air Pollution Control.

4.1.1 State chemical recovery
4.1.2 State the black liquor and its properties
4.1.3 Describe the chemical composition of black liquor
4.1.4 Describe distillation and oxidation of black liquor
4.1.5 State the use of evaporators
4.1.6 State the working of principles of evaporators
4.1.7 List the types of evaporators short tube evaporator, long tube evaporators, falling film evaporator, forced in circulation evaporator ,cascade evaporator
4.1.8 State the method of feeding in evaporators such as multiple effect and single effect
4.1.9 Calculate the steam economy in evaporator
4.1.10 Explain the recovery boiler and furnace
4.1.11 State the different types of recovery boilers and furnaces
4.1.12 State the function and operations of recovery boilers such as Babcox and Wilcox, tomilson, lankashare waste heat boiler electrostatic precipitator
4.1.13 Explain the function of ID, Fan, FD fan

4.2.0 Understand about caustisizing

4.2.1 State the principles of caustisizing process

4.2.2 State the function and chemical reactions of caustisizing
4.2.3 State green liquor

4.2.4 Specify the reaction taking place during the process of black liquor and boiler and caustisizing plant
4.2.5 State the functions and structure of lime shaker, classifiers, mud washers a lime debarring dryers
4.2.6 State lime reburning sludge disposal
4.2.7 List the chemical used in caustisizing operation
4.2.8 Calculate the lime requirement in a lime reburning kiln
4.2.9 Discuss the properties, transpositions, methods of handling storage and effect of lime impurities of make up chemicals
4.2.10 Describe the conventional sulphate recovery systems
4.2.11 State zeolite process and dimener alisation process
4.2.12 Calculation of white liquor for cooking

4.3.0 State effluent treatment

4.3.1 State the need of effluent treatment
4.3.2 State chemical used in effluent plant- such as polyelectrolyle alum etc.
4.3.3 Discuss the various effluents coming out of pulp and paper mills – such as air pollution, steam pollution, solid waste
4.3.4 Describe COD, BOD, and colour

4.4.0 State the method of air pollution control and their principles

4.4.1 State the equality control system in pulp mills

CONTENT DETAILS

MODULE I

Introduction – stock preparation-
Defibration – Hydra Pulper – batch and continuous, Kneaders, Deflakers
Storage chests and agitators – different types of stock and machine chests – agitator assemblies batch and continuous process of stock preparation-Beating & Refining processes – Theory - Beating and refining – factors influencing beating and refining – freeness – Hydration – Fibrillation – and their effects on paper properties different types of beaters & refiners, distinguish between beater & refiners Hollander beater- conical refiner, narrow wide angle shallow angle refiners single and double disc Consistency controllers and stock proportioning – different types of consistency controllers – working principles.

MODULE II

Purpose of sizing – internal and surface sizing – sizing chemicals such as rosin, alum, and wax their preparation, theory of sizing – parameters influencing sizing , acidic alkaline and Neutral Sizing
Surface sizing – Sizing chemicals starch as a sizing agent
Size presses - straight horizontal, inclined and speed sizes
Dyeing: Pigment dyes, Basic dyes, Acid dyes, Direct dyes, General Properties of different groups of dyes – different ways of adding the dye fastness of dye penetration of dyes into the fiber
Factor affecting stock dyeing – Beating and refining - sizing – fillers, pH, order of addition – pulp characteristics Filling & Loadings: - Purpose of filling and loadings – different types of fillers, such as

clay, talc, titanium dioxide calcium carbonate and magnesium carbonate.
Retention and effects of fillers on paper Properties – theory of retention – filtration. Co-flocculation – mechanical attachment Wet strength additives: Chemistry of additives influences of additives on cellulose – different types of additives – natural and synthetic preparation and addition of wet strength additives ,Basis weight, caliper, density, bulk, folding endurance, tear strength, burst strength, tensile,

stretch,paper testing – sample preparations surface properties – Roughness, smoothness pick strength, abrasion resistance oil absorbency, brightness, opacity, gloss sizing degree, colour, water absorbency - stock proportioning – different methods and types
Cleaning and screening – settling, centrifugal screens – HD and LD cleaning – Different methods of cleaning – pressure screens – centrifugal & cleaners – sorters.
Dearation – purpose – common type of dearators – decculators.
Head box – head box and its parts – distribution section, middle section- slice straight, convergent and divergent type of head box.
Type of head boxes- open and closed – role of distributor roll – high turbulence head box – different type of manifolds – head box over.
Flow, head box re circulation, jet wire ratio, head calculations (simple problems only).
Wire part – General arrangement and description of paper machine system and sheet formation typeof paper machine wire part – cylinder mould- fourdrinier – twin wire formers – special former Combinations.
Brief descriptions of different components of wire part foaming medium wire weave pattern sheet transfer factors affecting drainage and sheet formation.

MODULE III

(a) Drying – principles of drying different methods of drying conduction convection and radiation pre drying factors- factors affecting drying – heat content temperature moisture different types of dryers and its arrangement yankee cylinder (MG). Multi cylinder (MF) combined – infrared radiator, flaket use of air in paper drying.
Steam and condensate removal – cascade – pressure regulated differential pressure regulated -– ejectors – Thermo compressors- Condensate removal method – bucket, scoop siphon, rotating and stationary. Rimming calculation of heat transfer, heat transfer rate.
Overall heat transfer coefficient – simple problems
Dryer hoods open – closed high velocity exhaust fans, dryer felts and screens – its cleaning and Drying- Common troubles in drying – psychrometry.
Coating, calendering, reeling
Coating – coating pigments – binders – coating formulations – coating equipment's, air knife coater, brush coater, cast coating, flexible blade cotter, gravure coating, knife coating reverse roll coater – transfer roll coater – curtain coating
calendering – theory of calendering – effect of calendering on paper quality open side closed side calender swimming rolls – Importance of moisture on paper during calendaring common calender troubles and defects- hard and soft spots- sheet blackening cuts.
Finishing –sheet cutting rotary simplex duplex guillotines sorting counting packing –sheets – reels – roll.

wrapping machine weighing and labeling

reeling- pope reel – core mind reels – friction or surface drive reels converting – specialty – papers – finishing – winding different types of winder converting slitting.

MODULE IV

Chemical recovery
Black liquor – properties of black liquor – chemical composition of black liquor – deslication and Oxidation.
Evaporators – working principles of evaporator, types of evaporators- long tube evaporators – short tube evaporator – falling film evaporator- forced circulation evaporators – multiple effects single effects, cascade evaporators. Disc evaporators.
Steam economy calculation in evaporators – recovery boiler and furnace
Types - function, electrostatic, precipitator (ESP) reactions in recovery in boiler, make up chemical ID fan, FD fan, Boiler feed water properties.
Caustisizing
Functions and chemical reaction batch and continuous process – green liquor classifiers and sledge removal.
Green liquor, white liquor, sledge, sulphidity, caustisizing lime slakes, classifiers, white liquor clarifier mud washer operation techniques – lime reburning sludge disposal – make up chemicals – properties storage, transportation methods, handling and effect of lime impurities
Brief discussion of conventional sulphate recovery systems.
Zeolite process, demineralization process, simple calculation of white liquor for cooking.
Effluent treatment
Pulp and paper mill effluent – physical and chemical nature of effluent and its treatment pollution control board and its norms chemicals used in effluent treatment, air pollution and its control methods.

TEXT BOOKS

Ronald.G.Macdibak - Pulp & Paper Science, Vol. I, II, III - McGraw Hill Book Co., NewYork

James. P.Casey - Pulp & Paper Chemistry and Chemical Technology - Vol. I, II, III, IV

Tappi press - The Paper Machine wet press manual

REFERENCES

K.W. Britt - Hand Book o f Pulp & Paper Technology

G.A. Smook - Hand book of Pulp & Paper Technology

H.E Rance - Handbook of Paper Science, Vol. I & II